

TABLEAU COURSEWORK REPORT

Spotify 2023 — Exploratory Data Analysis & Dashboard Development

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1. Introduction

This report presents an exploratory data analysis of the **Spotify 2023 dataset** using Tableau. The objective of the analysis is to investigate patterns in song popularity, streaming performance, audio characteristics, artist contribution, platform exposure, and cross-platform presence.

Tableau was used to transform the raw dataset into interactive visualisations and dashboards. The analysis is divided into different analytical areas, allowing the user to explore individual aspects of the dataset as well as view the overall findings through a consolidated dashboard.

The final Tableau workbook focuses on converting raw data into meaningful visual insights through charts, KPI cards, filters, comparisons, and interactive dashboards.

2. Aim

The aim of this coursework is to **analyse the Spotify 2023 dataset using Tableau and develop interactive visualisations that reveal meaningful patterns in music streaming and platform performance.**

3. Objectives

The main objectives of the analysis are:

- To analyse the streaming performance of songs.
- To identify the most-streamed songs and artists.
- To study the audio characteristics of popular songs.

- To examine the relationship between platform exposure and streams.
 - To analyse the contribution of artists and collaborations.
 - To identify trends in music releases and streaming over time.
 - To compare the presence of songs across different music platforms.
 - To create interactive dashboards for effective data exploration.
 - To communicate the major findings through a consolidated Tableau dashboard.
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4. Dataset Description

The dataset used for this analysis is **Spotify 2023.xlsx**, containing information about songs that were prominent on Spotify during 2023.

The dataset contains information related to:

Category	Examples
Song Information	Track name, artist name
Streaming Performance	Streams
Release Information	Release date/year
Audio Characteristics	Danceability, Energy, Valence, Acousticness, Liveness, Speechiness, Instrumentalness
Platform Exposure	Playlist and chart-related measures
Artist Information	Artist and collaboration information

The different variables allow the dataset to be examined from multiple perspectives rather than relying only on the number of streams.

5. Tools and Technologies Used

Tableau

Tableau was the primary visualisation and analytical tool used for this coursework.

The following Tableau features were used:

- Dimensions and Measures

- Filters
- Calculated fields
- KPI cards
- Bar charts
- Line charts
- Scatter plots
- Heatmaps
- Dashboard containers
- Interactive dashboard filters
- Story/dashboard navigation

Microsoft Excel

The original dataset was provided in Excel format and imported into Tableau for analysis.

6. Methodology

The analysis followed a structured data visualisation process.

Step 1 — Data Connection

The **Spotify 2023 Excel workbook** was imported into Tableau.

Step 2 — Data Understanding

The available dimensions and measures were examined to understand the structure of the dataset and determine which fields were suitable for each analysis.

Step 3 — Worksheet Creation

Individual worksheets were created for specific analytical questions. Different visualisation types were selected according to the nature of the data.

Step 4 — Dashboard Development

Related worksheets were combined into dashboards. Filters, KPI cards and interactive elements were added to make the dashboards easier to explore.

Step 5 — Visual Formatting

Titles, labels, legends, spacing and dashboard layouts were adjusted to improve readability and maintain consistency.

Step 6 — Story Development

7. Dashboard 1 — Music DNA

7.1 Purpose

The **Music DNA** dashboard focuses on the audio characteristics and streaming performance of songs in the dataset.

The purpose of this dashboard is to understand what the musical profile of the songs looks like and how the audio characteristics can be examined alongside streaming performance.

Insert Screenshot Here

[INSERT DASHBOARD 1 SCREENSHOT]

7.2 Visualisations Used

The dashboard includes visualisations such as:

- Audio feature analysis
- Top-streamed songs
- KPI cards
- Comparisons of musical characteristics

Audio characteristics such as **danceability, energy, valence, acousticness, instrumentalness, liveness and speechiness** provide additional context about the songs.

7.3 Analysis

The dashboard allows the user to examine whether highly streamed songs exhibit noticeable patterns in their audio characteristics.

The top-song visualisation highlights the tracks that achieved the highest streaming numbers, while the audio-feature visualisation provides a broader understanding of the musical characteristics present within the dataset.

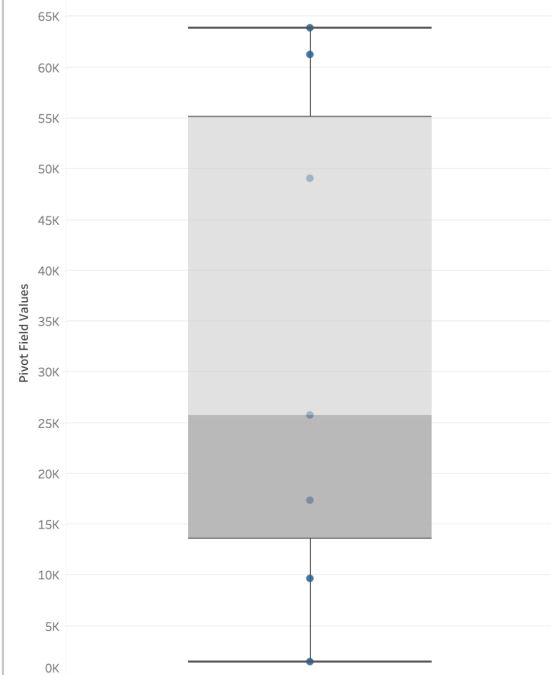
The combination of these visualisations makes it possible to compare **song popularity with musical characteristics** rather than analysing streaming performance alone.

7.4 Insight

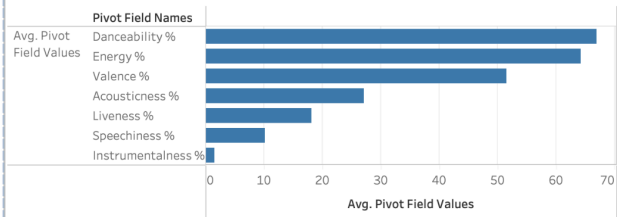
The dashboard demonstrates that streaming performance can be analysed from multiple dimensions. Audio features provide additional context for understanding popular songs, although the visual relationships should be interpreted as descriptive associations rather than evidence of causation.

01 - MUSIC DNA

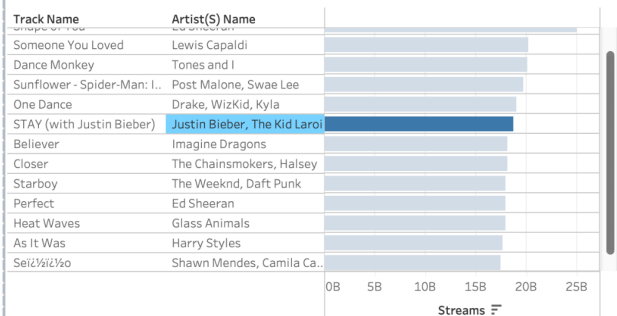
03 - Audio Feature Distribution



Average Audio Profile of Popular Songs



Top 15 Most-Streamed Songs



8. Dashboard 2 — Platform Wars

8.1 Purpose

The **Platform Wars** dashboard examines the relationship between platform exposure and song streaming performance.

The objective is to understand how playlist and chart presence across platforms can be compared with the number of streams received by songs.

Insert Screenshot Here

[INSERT DASHBOARD 2 SCREENSHOT]

8.2 Visualisations Used

The dashboard includes:

- Platform playlist comparison
- Platform chart comparison
- Platform exposure analysis
- Streaming performance comparison

8.3 Analysis

Platform exposure can provide an indication of how visible a song is to listeners.

The visualisations allow platform-related measures to be compared with streaming performance. This makes it possible to identify patterns between songs receiving greater platform exposure and their overall streaming numbers.

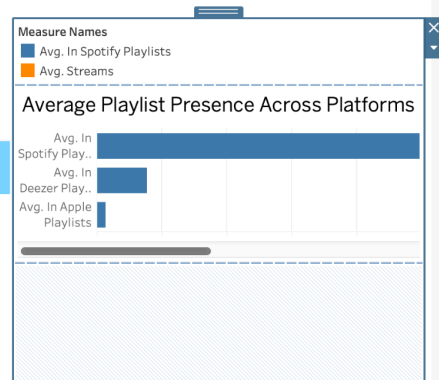
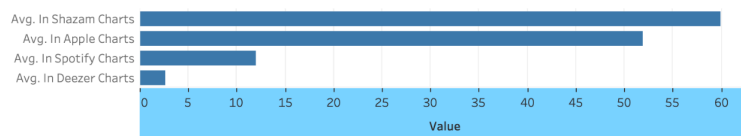
8.4 Insight

The analysis provides a visual comparison between **visibility and streaming performance**. Songs with greater platform exposure can be examined alongside their streaming numbers to identify potential associations.

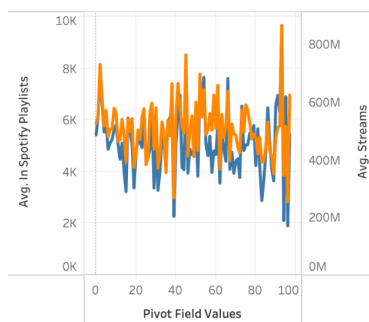
However, platform exposure and streams should not automatically be interpreted as a cause-and-effect relationship because the dataset is observational.

02 - PLATFORM COMPARISON

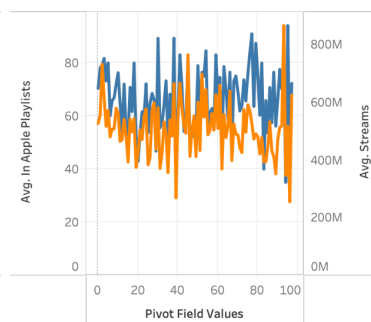
Average Chart Presence Across Platforms



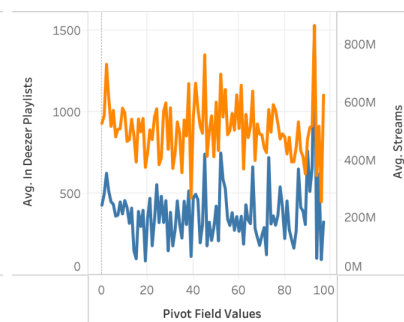
Spotify Playlist vs Streams



Apple Playlist vs Streams



Deezer Playlist vs Streams



9. Dashboard 3 — Artist Impact

9.1 Purpose

The **Artist Impact** dashboard investigates artist-level performance and the role of collaborations in streaming.

The dashboard focuses on identifying prominent artists and examining whether the number of artists associated with a track is related to its streaming performance.

Insert Screenshot Here

[INSERT DASHBOARD 3 SCREENSHOT]

9.2 Visualisations Used

The dashboard includes:

- Top artists
- Artist-wise streaming analysis
- Artist count versus streams
- Solo versus collaboration comparison

9.3 Analysis

The top-artist visualisation highlights artists that are strongly represented among the songs in the dataset.

The artist-count analysis examines whether songs featuring multiple artists display different streaming patterns from songs performed by a single artist.

The solo-versus-collaboration comparison provides another perspective on the relationship between artist participation and song performance.

9.4 Insight

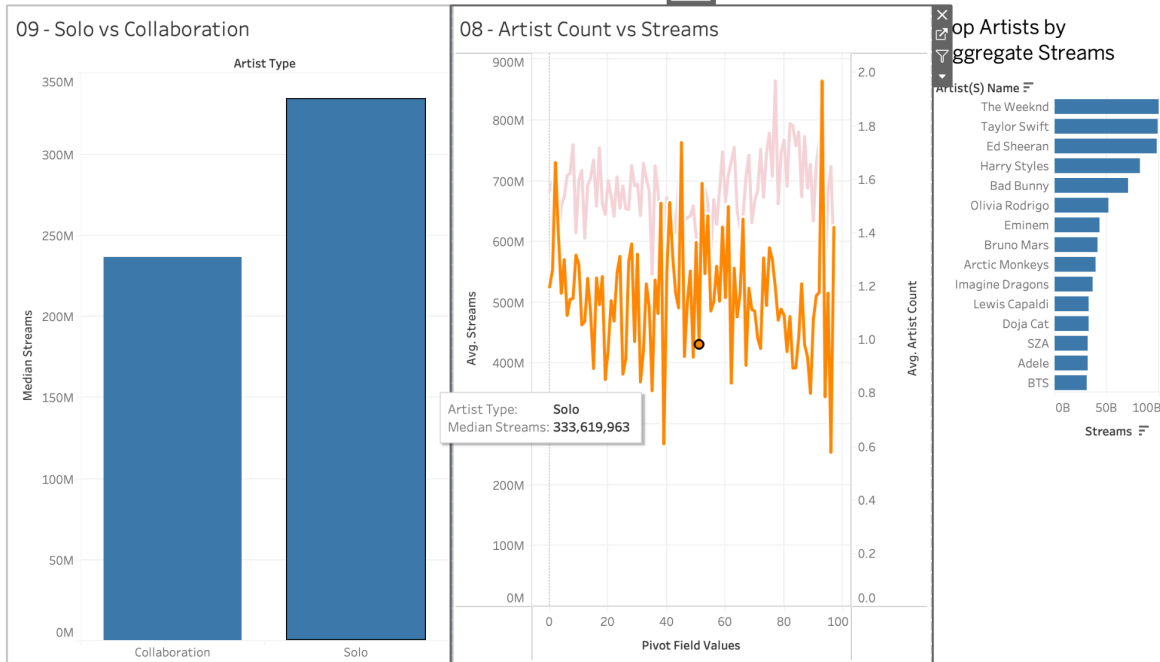
Artist contribution is an important dimension when analysing music streaming data. The dashboard demonstrates how both individual artist presence and collaborations can be examined alongside streaming performance.

The results describe patterns within the dataset and do not establish that collaborations directly cause higher or lower streams.

03 - ARTIST IMPACT

Measure Names

- Avg. Artist Count
- Avg. Streams



10. Dashboard 4 — Music Through Time

10.1 Purpose

The **Music Through Time** dashboard analyses the temporal characteristics of the dataset.

The objective is to understand how releases and streaming performance vary across time.

Insert Screenshot Here

[INSERT DASHBOARD 4 SCREENSHOT]

10.2 Visualisations Used

The dashboard includes:

- Release trends
- Streaming trends
- Time-based comparisons
- Audio-feature trends where applicable

10.3 Analysis

Time-based visualisations allow changes in music releases and streaming performance to be observed across the available period.

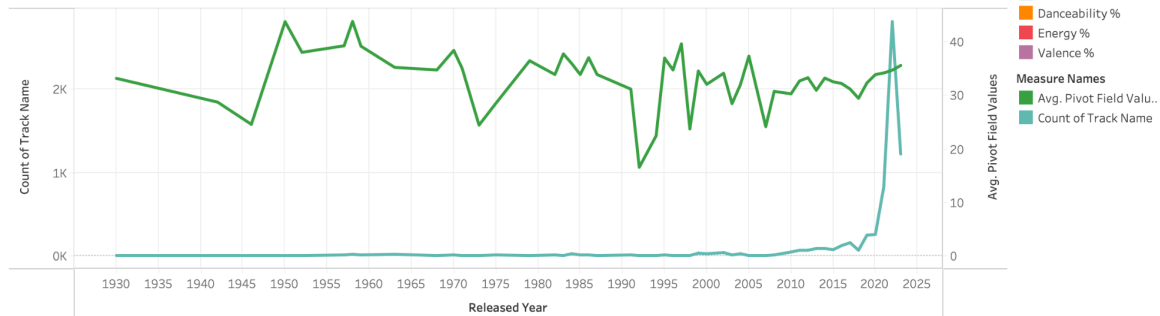
The trend charts make it easier to identify periods of increased activity and compare streaming performance across different time intervals.

10.4 Insight

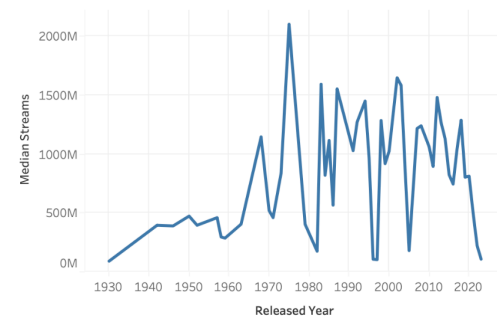
The temporal analysis provides a dynamic perspective of the dataset and demonstrates how music trends can change over time rather than remaining constant.

04 - MUSIC THROUGH TIME

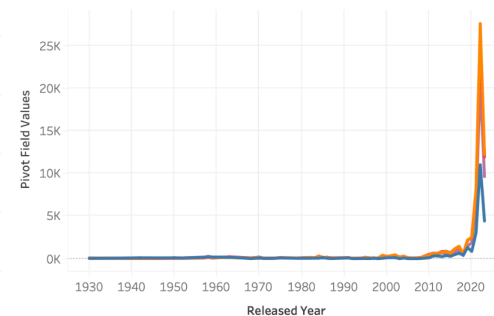
11 - Releases Over Time



14 - Streaming Trend by Release Year



13 - Audio Trends Over Time



11. Dashboard 5 — Cross-Platform Reach

11.1 Purpose

The **Cross-Platform Reach** dashboard examines how songs are represented across different music platforms.

Instead of focusing exclusively on Spotify, this analysis considers the broader platform presence of songs.

Insert Screenshot Here

[INSERT DASHBOARD 5 SCREENSHOT]

11.2 Visualisations Used

The dashboard includes:

- Cross-platform reach
- Platform heatmap
- Spotify versus Apple comparison
- Platform presence analysis

11.3 Analysis

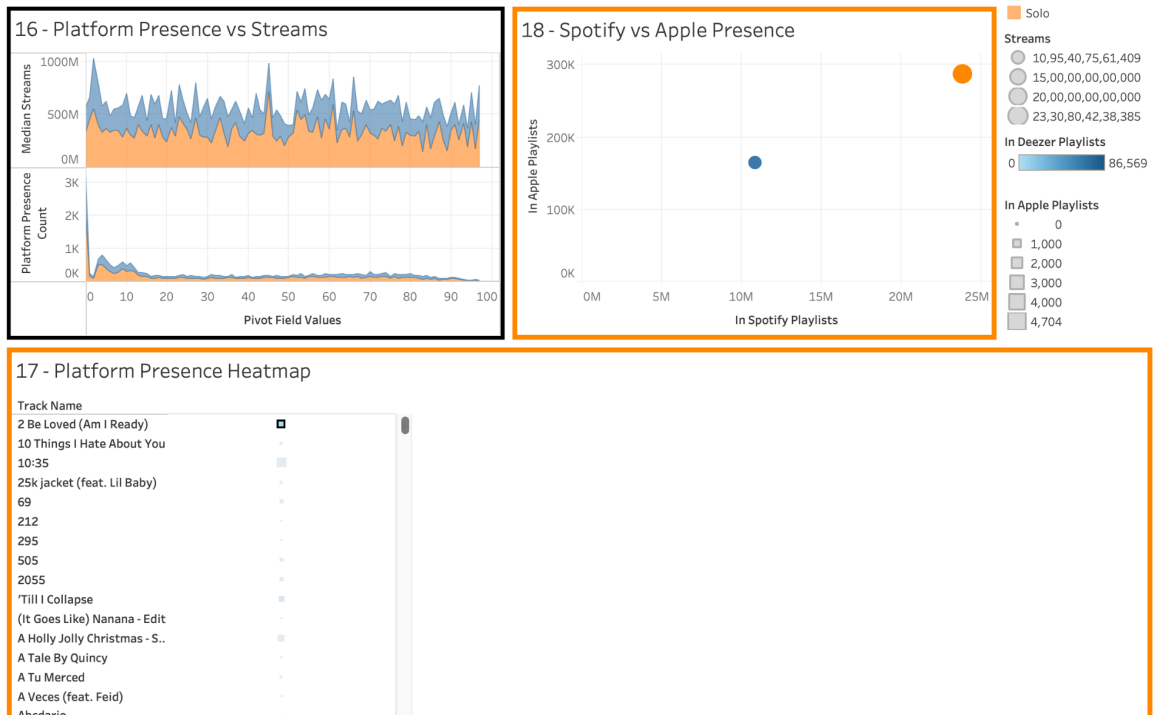
The cross-platform analysis allows songs to be compared according to their presence across different platforms.

The heatmap provides a compact way of identifying differences in platform exposure, while the Spotify-versus-Apple comparison provides a direct view of platform presence.

11.4 Insight

Cross-platform analysis provides a broader understanding of song reach. Some songs may have strong representation across several platforms, while others may have greater visibility on particular platforms.

05 - CROSS-PLATFORM REACH



12. Consolidated Insights Dashboard

12.1 Purpose

The final dashboard brings together the most important findings from the individual analyses.

It provides a high-level overview of the Spotify 2023 dataset using selected KPIs and visualisations.

Insert Screenshot Here

[INSERT FINAL DASHBOARD SCREENSHOT]

12.2 Analysis

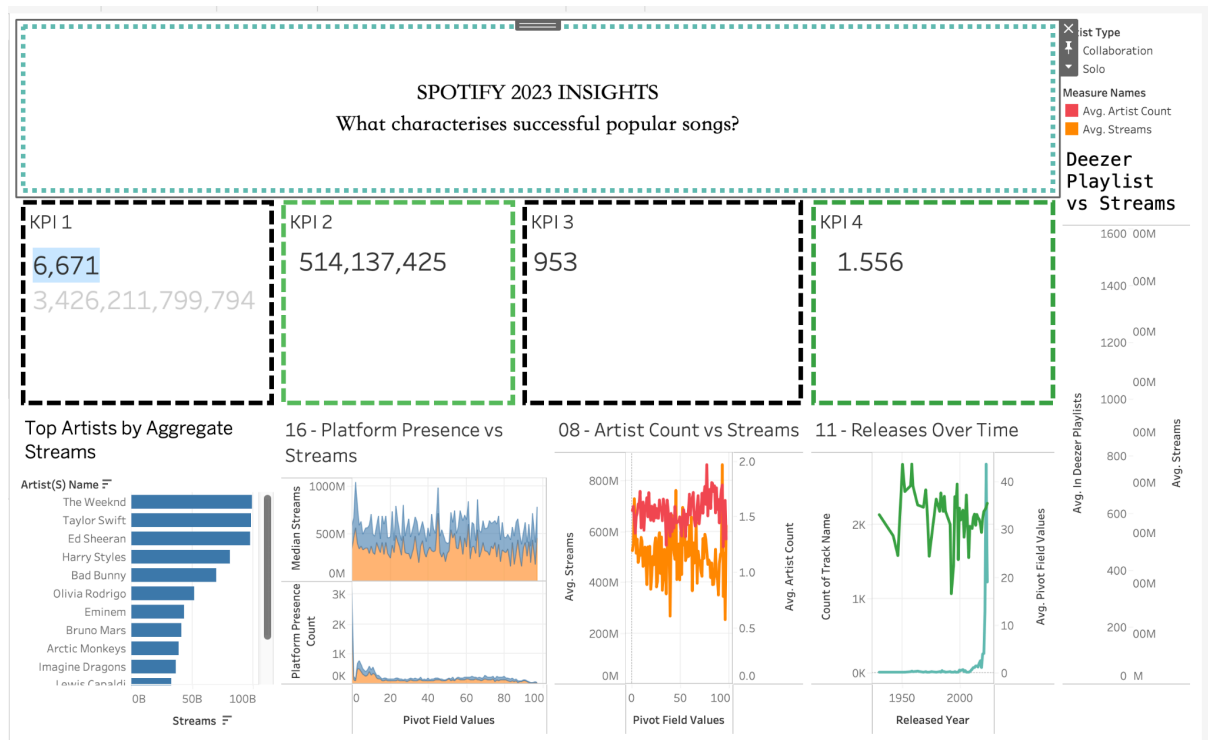
The consolidated dashboard allows users to quickly understand the major patterns identified throughout the analysis.

It combines information related to:

- Streaming performance
- Popular songs

- Artists
- Platform exposure
- Audio characteristics
- Cross-platform presence
- Trends over time

The dashboard acts as an entry point for users who want to obtain an overview before exploring the individual analytical dashboards in greater detail.



13. Key Findings

The analysis of the Spotify 2023 dataset produced several important observations:

1. **Streaming performance varies considerably between songs**, making ranking and comparison useful for identifying the most prominent tracks.
2. **Audio characteristics provide additional context** when studying highly streamed songs and allow song popularity to be examined beyond simple stream counts.
3. **Platform exposure can be compared with streaming performance** to identify patterns between visibility and popularity.
4. **Artist-level analysis provides another important perspective**, particularly when comparing individual artists and collaborative tracks.
5. **Time-based analysis helps identify changes in music activity and streaming performance** across the available period.

6. **Cross-platform analysis provides a broader perspective of song reach** than analysing Spotify alone.
 7. **Interactive Tableau dashboards make the analysis easier to explore**, allowing users to filter and compare different aspects of the dataset.
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14. Overall Analysis

The complete Tableau workbook demonstrates how a large music dataset can be converted into an interactive analytical solution.

Rather than focusing on one metric, the analysis combines several dimensions including **streams, artists, audio features, time and platform exposure**.

The dashboards complement one another:

- **Music DNA** focuses on the characteristics of songs.
- **Platform Wars** focuses on platform exposure.
- **Artist Impact** focuses on artists and collaborations.
- **Music Through Time** focuses on temporal patterns.
- **Cross-Platform Reach** focuses on broader platform presence.
- **Spotify 2023 Insights** brings the major observations together.

This structure creates a logical progression from individual song characteristics to broader patterns in the music streaming ecosystem.

15. Conclusion

The Tableau analysis successfully transforms the Spotify 2023 dataset into a structured and interactive visual analytics solution.

The analysis demonstrates how Tableau can be used to explore large datasets through multiple perspectives, including **streaming performance, audio characteristics, artist contribution, platform exposure, temporal patterns and cross-platform reach**.

The use of dashboards and interactive filters makes the analysis more accessible and allows users to investigate specific patterns according to their interests.

Overall, the project demonstrates the importance of combining **data preparation, analytical thinking and effective visualisation** to transform raw data into meaningful insights.

16. Limitations

The analysis has several limitations:

- The analysis is restricted to the variables available in the provided Spotify 2023 dataset.
 - The dataset represents a specific period and therefore may not describe long-term music industry trends.
 - Relationships observed in the visualisations represent associations and do not necessarily indicate causation.
 - Streaming data can be highly uneven, with a relatively small number of songs potentially accounting for a large proportion of total streams.
 - Platform metrics may not be directly comparable because different platforms use different systems for playlists, charts and exposure.
 - Missing or inconsistent values in the original dataset may affect certain comparisons.
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17. Future Scope

The analysis can be extended in several ways:

- Include Spotify data from multiple years to study long-term trends.
 - Add genre and language information.
 - Include geographic or regional listening data.
 - Analyse monthly or weekly streaming trends.
 - Compare additional music platforms.
 - Use statistical techniques to test relationships identified during the exploratory analysis.
 - Develop predictive models for estimating future song performance.
 - Add more advanced Tableau parameters and interactive controls.
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18. Final Summary

The Spotify 2023 Tableau project demonstrates a complete **data-to-dashboard workflow**. The raw dataset was transformed into multiple analytical views and organised into an interactive dashboard structure.

The analysis explores music from different perspectives, including **what songs are popular, what their musical characteristics are, how artists contribute to performance, how platform exposure relates to streams, how patterns change over time, and how songs reach different platforms**.

The final Tableau workbook therefore provides both detailed analysis and a consolidated overview of the Spotify 2023 dataset.